A finite-state proof of the A181280 formula

Ordered binary matrices with reverse-ordered Gram rows

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Abstract

We prove the six-term formula of Kauers and Koutschan for the number of binary $4 \times n$ matrices whose rows increase strictly in lexicographic order while the rows of their Gram matrix over $\mathbb{F}_2$ decrease strictly. A column automaton records three comparisons and ten Gram parities. Fourier transformation on the parity group decomposes its transition operator into 1,024 matrices of size eight. Quadratic character sums and a four-level filtration give a universal monic annihilator of degree 16. Consequently, sixteen exact initial values imply the formula at every length. We supply the full finite certificate as a short, executable integer-arithmetic program, an explicit rational generating function, and separately implemented checks. The result is a computer-assisted proof with an analytic all-length bridge; no formal proof-assistant verification or external peer review is claimed.

1 Problem and result

Compare binary words of equal length lexicographically, with $0 < 1$ and the leftmost coordinate compared first. Let $a(n)$ count the matrices $M \in \mathbb{F}_2^{4\times n}$ with rows $r_1 < r_2 < r_3 < r_4$ such that the rows g_1, g_2, g_3, g_4 of $G = MM^{\mathsf{T}}$ satisfy $g_1 > g_2 > g_3 > g_4$. Gram multiplication is over $\mathbb{F}_2$; counting is over the ordinary integers. Columns remain ordered and labeled by position. For convenience define $a(0) = 0$, consistent with the impossibility of four distinct empty rows.

Theorem 1. *We have $a(1) = a(2) = a(3) = 0$. For every integer $n \geq 4$,*

$$a(n) = \frac{2^{2n-11}}{3}(6n^2 - 219n + 820) - \frac{2^{n-5}}{9}(3n + 32) - \frac{113}{3}(-1)^n 2^{3n-14} + 2^{4n-9} - \frac{(-1)^n 2^{2n-11}}{3}(13n - 164) + \frac{2^{3n-14}}{9}(288n - 3473). \tag{1}$$

This is precisely Conjecture 20 in Section 6.5 of Kauers–Koutschan [1], and the conjectural formula in OEIS A181280 [2]. The four initial values, including $n = 0$, are treated separately: the right side of (1) does not give them. For a first example, the reverse identity matrix

$$M = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \quad \text{has} \quad MM^{\mathsf{T}} = I_4.$$

Its rows increase and the rows of I_4 decrease, so it is one of the 58 matrices counted by $a(4)$.

Literature boundary. The latest arXiv version located in the source audit was v2 of 24 April 2023; the published article appeared in the *Journal of Integer Sequences*, volume 26, Article 23.4.5, on that date. Its Conjecture 20 is the target above. The OEIS entry retrieved on 25 October 2026 still marks the formula conjectural (entry revision 11, dated 7 March 2023). A bounded search for later proofs found no equivalent proof. This records the search boundary, not a claim of exhaustive priority. The formula itself is credited to Kauers and Koutschan. Source URLs, snapshots, and audit details are included in the accompanying literature ledger.

2 An exact column automaton

Write $\mathcal{G} = \mathbb{F}_2^{10}$ for the additive group of symmetric 4×4 binary matrices, represented by their upper-triangular entries in the order

$$(1,1),(1,2),(1,3),(1,4),(2,2),(2,3),(2,4),(3,3),(3,4),(4,4).$$

The state is (S, g), where $S \subseteq \{1,2,3\}$ records which adjacent row comparisons have already become strictly increasing, and $g \in \mathcal{G}$ is the accumulated Gram matrix. Initially $(S, g) = (\varnothing, 0)$.

For a new column $c = (c_1, c_2, c_3, c_4)^{\mathsf{T}} \in \mathbb{F}_2^4$, the transition is forbidden if some $i \notin S$ has $(c_i, c_{i+1}) = (1, 0)$. Otherwise define

$$S' = S \cup \{i \notin S : (c_i, c_{i+1}) = (0,1)\}, \qquad g' = g + cc^{\mathsf{T}}. \tag{2}$$

Addition in the second coordinate is modulo two. An accepting state has $S = \{1,2,3\}$ and strictly decreasing rows in g.

Lemma 2 (Counting bijection)**.** *The length-n accepting column paths of this automaton are in bijection with the matrices counted by $a(n)$. In particular, if T is its transition operator on a rational vector space of dimension $8 \cdot 1024 = 8192$, v is the initial-state vector, and ℓ sums accepting-state coordinates, then*

$$a(n) = \ell(T^n v). \tag{3}$$

Proof. Until adjacent row prefixes first differ, their comparison is unresolved. Their first unequal bits must be $(0,1)$; a first pair $(1,0)$ irreversibly violates the required order. After the first $(0,1)$, subsequent columns cannot change the lexicographic comparison. Thus (2) records exactly the adjacent comparisons, and all three must be resolved for strict row order. Adjacent strict comparisons imply strict order of all four rows by transitivity. Also $MM^{\mathsf{T}} = \sum_{j=1}^{n} c^{(j)}(c^{(j)})^{\mathsf{T}}$ over $\mathbb{F}_2$, giving exactly the parity update. The terminal test is the required Gram-row condition. The sequence of columns determines one matrix uniquely and every matrix determines one sequence. If distinct columns give the same state transition, their contributions are added as separate edges; no multiplicity is discarded. □

3 Fourier reduction and a universal annihilator

For $h, g \in \mathcal{G}$, use the upper-triangular coordinate pairing

$$\langle h, g \rangle = \sum_{1 \le i \le j \le 4} h_{ij} g_{ij} \in \mathbb{F}_2, \qquad \chi_h(g) = (-1)^{\langle h, g \rangle}.$$

This is a coordinate pairing, not the matrix trace pairing, which would lose off-diagonal information in characteristic two. For each h, let $w_{S,h} = \sum_{g \in \mathcal{G}} \chi_h(g) e_{S,g}$, where $e_{S,g}$ is a state basis vector. Character orthogonality gives

$$\sum_{g \in \mathcal{G}} \chi_h(g) \chi_{h'}(g) = 1024\, \mathbf{1}_{h=h'}.$$

Consequently these vectors form a basis and decompose the space into 1,024 invariant subspaces, each spanned by eight $w_{S,h}$.

Indeed, translating the Gram state by cc^{T} multiplies $w_{S,h}$ by $\chi_h(cc^{\mathsf{T}})$. The block operator T_h has the explicit action

$$T_h w_{S,h} = \sum_{c \text{ allowed at } S} (-1)^{\sum_{i \le j} h_{ij} c_i c_j} w_{S'(S,c),h}. \tag{4}$$

Since $S' \supseteq S$, this operator is triangular in any ordering extending inclusion of subsets. Only its diagonal and its four subset-size levels will be needed.

Lemma 3 (Quadratic character sums)**.** *If $q : \mathbb{F}_2^b \to \mathbb{F}_2$ is a polynomial function of degree at most two with $q(0) = 0$, the sum $W(q) = \sum_x (-1)^{q(x)}$ belongs to the following sets:*

b	*Possible values (a sufficient list)*
1	$0, 2$
2	$-2, 0, 2, 4$
3	$-4, 0, 4, 8$
4	$-8, -4, 0, 4, 8, 16$

Proof. The polar form $B(x, z) = q(x + z) + q(x) + q(z)$ is alternating and bilinear. Its rank r is even. For completeness, if B is nonzero, choose u, v with $B(u, v) = 1$. Their two-dimensional span is nondegenerate, and splitting off its orthogonal complement reduces the rank by two; induction proves evenness. Let R be the radical, of dimension $b - r$. Changing variables from (x, y) to $(x, z = x + y)$ gives

$$W(q)^2 = \sum_z (-1)^{q(z)} \sum_x (-1)^{B(x,z)} = 2^b \sum_{z \in R} (-1)^{q(z)}.$$

The restriction of q to R is linear, so its sum is either zero or $|R|$. Therefore $W(q) = 0$ or $W(q) = \pm 2^{b-r/2}$. If $r = 0$, q is linear on the whole space, giving only 0 or 2^b, never -2^b. The possible even ranks for $b = 1, 2, 3, 4$ give the displayed lists. □

Lemma 4 (Diagonal and filtration)**.** *For every $h \in \mathcal{G}$, the operator T_h is annihilated by*

$$P(X) = X^4(X - 2)^2(X + 2)(X - 4)^3(X + 4)^2(X - 8)^2(X + 8)(X - 16). \tag{5}$$

Proof. A column leaves S unchanged precisely when $c_i = c_{i+1}$ for all $i \notin S$. The cuts at the indices of S split the four row positions into $b = |S| + 1$ nonempty consecutive blocks. Such a column is constant on each block, and all 2^b choices of block values are allowed. Substituting these values in $\sum_{i \le j} h_{ij} c_i c_j$ produces a quadratic polynomial with zero constant term. Hence the diagonal coefficient at S lies in the bth list of Lemma 3.

Let $p_b(X)$ be the product of $X - \lambda$ over that list; explicitly,

$$p_1 = X(X - 2), \qquad p_2 = X(X + 2)(X - 2)(X - 4),$$
$$p_3 = X(X + 4)(X - 4)(X - 8), \qquad p_4 = X(X + 8)(X + 4)(X - 4)(X - 8)(X - 16).$$

For $k = 0, \dots, 4$, let V_k be the span of the basis vectors with $|S| \ge k$; in particular V_0 is the block space and $V_4 = 0$. Each V_k is invariant. On V_k / V_{k+1} the operator is diagonal: there are no transitions between distinct subsets of the same size. All its diagonal entries are roots of p_{k+1}, so

$$p_{k+1}(T_h) V_k \subseteq V_{k+1}.$$

Applying $p_1(T_h), p_2(T_h), p_3(T_h), p_4(T_h)$ in order sends V_0 into $V_4 = 0$. Their product equals $P(T_h)$ because polynomials in one operator commute. Multiplying the displayed factors yields exactly (5). □

Proposition 5 (All-length recurrence)**.** *Let E denote the forward shift, $(\mathsf{E}u)(n) = u(n+1)$. Then $P(\mathsf{E})a = 0$ for every $n \ge 0$. Thus the sixteen values $a(0), \dots, a(15)$ determine the full sequence.*

Proof. The character basis decomposes T into the T_h, each annihilated by P, so $P(T) = 0$. Equation (3) now gives $\sum_{j=0}^{16} P_j a(n + j) = \ell(T^n P(T) v) = 0$. The coefficient of X^{16} in P is one. This recurrence therefore determines $a(n + 16)$ from preceding terms for every $n \ge 0$. No assertion that P is minimal is needed. □

4 Exact initial values and proof of the formula

Let $F(n)$ be the right side of (1), interpreted over the rationals for all $n \geq 0$, and let $b(n) = 0$ for $0 \leq n < 4$ and $b(n) = F(n)$ for $n \geq 4$. We next show that b satisfies the same universal recurrence before comparing initial values.

Lemma 6. *The sequence b satisfies $P(\mathsf{E})b = 0$ for every $n \geq 0$.*

Proof. Each summand of F is a polynomial in n times λ^n. The six bases and polynomial degrees are

$$(\lambda, d) = (4, 2), (2, 1), (-8, 0), (16, 0), (-4, 1), (8, 1).$$

For a polynomial f of degree d, $(\mathsf{E} - \lambda)(\lambda^n f(n)) = \lambda^{n+1}(f(n+1) - f(n))$; iterating $d+1$ times gives zero. It follows that $Q(\mathsf{E})F = 0$, where

$$Q(X) = (X-2)^2(X-4)^3(X+4)^2(X-8)^2(X+8)(X-16). \tag{6}$$

The correction $b - F$ is supported at $n = 0, 1, 2, 3$, hence $\mathsf{E}^4(b - F) = 0$ on nonnegative indices. As $P(X) = X^4(X+2)Q(X)$, both F and $b - F$ are annihilated by $P(\mathsf{E})$. This explicitly accounts for the transient prefix. □

The finite count is computed by the following exact recurrence, which is simply the automaton written in coordinates:

$$d_0(S, g) = \mathbf{1}_{S=\varnothing, g=0}, \qquad d_{m+1}(S', g') = \sum_{\substack{S,g,c \text{ allowed} \\ S'(S,c)=S',\ g+cc^{\mathsf{T}}=g'}} d_m(S, g). \tag{7}$$

Summing $d_m(\{1, 2, 3\}, g)$ over the decreasing-row Gram matrices gives Table 1. The program in Appendix A supplies the complete executable finite certificate and checks equality with b. It uses arbitrary-precision integers and exact rational arithmetic. The table is independently reproduced by the Fourier and interval partition implementations in the supplement.

n	$a(n) = b(n)$	n	$a(n) = b(n)$
0	0	8	8 211 776
1	0	9	133 693 904
2	0	10	2 140 571 200
3	0	11	34 361 115 072
4	58	12	549 587 348 992
5	1 629	13	8 798 356 254 976
6	28 924	14	140 744 002 571 264
7	507 052	15	2 252 082 614 856 704

Table 1: The complete set of initial values needed for the degree-16 recurrence.

Proof of Theorem 1. Proposition 5 and Lemma 6 imply $P(\mathsf{E})(a - b) = 0$. The exact calculation in Table 1 gives $(a - b)(0) = \cdots = (a - b)(15) = 0$. Since P is monic of degree 16, induction on the recurrence gives $(a - b)(n) = 0$ for all $n \geq 0$. The definition of b is precisely the theorem, including the three exceptional positive indices. □

Remark 7. The values of $F(n)$ at $n = 0, 1, 2, 3$ are respectively $103/4096, -71/1024, -1/64, -1/16$. Substituting the conjecture into a recurrence without correcting this prefix would prove a different sequence. The factors X^4 above make the all-length argument valid.

5 Generating function and consequences

Corollary 8. *With $A(z) = \sum_{n\ge 0} a(n) z^n$, one has $A(z) = N(z)/D(z)$, where*

$$\begin{aligned} D(z) &= (1-2z)^2(1-4z)^3(1+4z)^2(1-8z)^2(1+8z)(1-16z), \\ N(z) &= 58z^4 - 227z^5 - 9052z^6 + 68976z^7 - 16960z^8 \\ &\quad - 1025408z^9 + 3055104z^{10} - 3796992z^{11} + 2834432z^{12} \\ &\quad - 2490368z^{13} + 2097152z^{14}. \end{aligned}$$

This fraction is reduced. The minimal eventual constant-coefficient recurrence has characteristic polynomial Q in (6), of degree 11, and $Q(\mathsf{E})a(n) = 0$ for every $n \ge 4$.

Proof. Theorem 1 and $Q(\mathsf{E})F = 0$ imply the claimed recurrence starting at $n = 4$. Since $D(z) = z^{11}Q(1/z)$, coefficients of $D(z)A(z)$ vanish at every degree at least 15. Multiplying its first fifteen coefficients using Table 1 gives the displayed N. To see that the fraction is reduced, recall $\sum_{n\ge 0} t^n = (1-t)^{-1}$ and apply $t\,d/dt$ repeatedly: a nonzero polynomial of degree d times λ^n has a generating function with a pole of exact order $d+1$ at $z = 1/\lambda$. Each of the six polynomial factors in F has its stated nonzero leading coefficient. Their poles are distinct; adding the finite prefix correction cannot cancel any pole. Thus all factors and multiplicities of D are necessary. Equivalently, every eventual constant-coefficient annihilator must contain Q. □

Corollary 9. *As $n \to \infty$,*

$$a(n) = \frac{16^n}{512} + O(n8^n), \qquad \frac{a(n)}{\binom{2^n}{4}} \longrightarrow \frac{3}{64}.$$

Proof. The remaining terms of (1) are $O(n8^n)$; in particular $n^2 4^n = O(n8^n)$. Choosing any four distinct length-n binary words determines exactly one increasingly ordered matrix, so the denominator counts all such matrices and is asymptotic to $16^n/24$. The ratio follows. □

6 Verification, reproducibility, and scope

The mathematical dependence of the proof is explicit: the column bijection gives (3); the quadratic-sum lemma and the filtration give $P(T) = 0$; the elementary exponential-polynomial calculation gives $P(\mathsf{E})b = 0$; and the finite certificate supplies the sixteen initial equalities. Only the last step needs a numerical calculation. It is exhaustive, finite, and exact, rather than statistical or based on a fitted recurrence.

The accompanying programs use Python 3.9 or newer and only its standard library. Run from the package root:

```
python src/paper_certificate.py
python src/verify.py --write
python src/fourier_proof.py --max-n 30 --check-all-vectors
python src/independent_check.py --output certificates/independent_check.json
```

The first program is printed in full below. The second computes the direct 8,192-state counts through $n = 30$ and checks the rational generating function coefficients. The third separately constructs the Fourier blocks and evaluates $P(T_h)$ at all eight basis vectors for all 1,024 characters. The fourth uses interval partitions and tuples of quadratic coefficients, imports no other project module, checks all 65,536 entries of the matrix identities, and directly enumerates the increasing row quadruples for $1 \le n \le 6$. The full Fourier counts agree with the direct counts on their common checked range. The extra terms beyond $n = 15$ are diagnostic checks; the proof of the tail is Proposition 5.

The manuscript, certificate, and implementations were produced by GPT-6 Astra, an AI system, including separate internal passes for literature checking, an alternative algebraic derivation, and independent code. The word "independent" refers to these implementations, not to independent human peer review. No Lean verification is claimed. The proof covers the entire stated four-row counting problem; it makes no classification claim for other numbers of rows and no claim that all increasingly ordered binary matrices obey the additional Gram-row restriction.

A Complete executable finite certificate

The ten Gram bits follow the upper-triangular order in Section 2. The three mask bits use adjacent row positions $0, 1, 2$ in Python's zero-based indexing. Encoding a row as a four-bit nonnegative integer with its first entry most significant implements the specified lexicographic order exactly. In the code, `g ^ h` is bitwise exclusive-or. Every loop has explicit finite bounds; no input file or saved count is read. The asserted list is independently recomputed before it is checked.

```
"""Self-contained finite certificate printed in the paper; Python 3.9+."""
from fractions import Fraction as F

pairs = [(i, j) for i in range(4) for j in range(i, 4)]
def gram_rows(g):
    rows = [0] * 4
    for k, (i, j) in enumerate(pairs):
        if (g >> k) & 1:
            rows[i] |= 1 << (3-j)
            rows[j] |= 1 << (3-i)
    return rows

good = []
for g in range(1024):
    r = gram_rows(g)
    if r[0] > r[1] > r[2] > r[3]:
        good.append(g)
assert len(good) == 48

edges = [[] for _ in range(8)]
for s in range(8):
    for c in range(16):
        b = [(c >> (3-i)) & 1 for i in range(4)]
        if any(not ((s >> i) & 1) and b[i] > b[i+1]
               for i in range(3)):
            continue
        t = s
        for i in range(3):
            if b[i] < b[i+1]:
                t |= 1 << i
        h = sum((b[i]*b[j]) << k
                for k, (i, j) in enumerate(pairs))
        edges[s].append((t, h))

v = [[0]*1024 for _ in range(8)]
v[0][0] = 1
counts = []
for n in range(16):
    counts.append(sum(v[7][g] for g in good))
    w = [[0]*1024 for _ in range(8)]
    for s in range(8):
        for t, h in edges[s]:
            for g, value in enumerate(v[s]):
                w[t][g ^ h] += value
    v = w

def formula(n):
    return (F(4**n*(6*n*n-219*n+820), 6144)
            - F(2**n*(3*n+32), 288)
            - F(113*(-8)**n, 49152) + F(16**n, 512)
            - F((-4)**n*(13*n-164), 6144)
```

```
            + F(8**n*(288*n-3473), 147456))

assert counts == [0, 0, 0, 0, 58, 1629, 28924, 507052,
    8211776, 133693904, 2140571200, 34361115072,
    549587348992, 8798356254976, 140744002571264,
    2252082614856704]
assert all(counts[n] == (0 if n < 4 else formula(n))
           for n in range(16))
print('PASS: exact initial values n=0..15; 48 Gram patterns')
```

B Explicit universal recurrence

For an additional arithmetic audit, expansion of (5) gives

$$\begin{aligned} P(X) = X^{16} - 30X^{15} + 180X^{14} + 2040X^{13} - 24288X^{12} \\ + 15360X^{11} + 624640X^{10} - 1812480X^{9} - 4349952X^{8} \\ + 22609920X^{7} - 7864320X^{6} - 62914560X^{5} + 67108864X^{4}. \end{aligned}$$

Replacing X^j by $a(n+j)$ gives a valid recurrence for all $n \geq 0$. The sparse prefix coefficients are zero, as required by the four-step transient. The recurrence used to prove the formula is this established operator identity, not a recurrence inferred from Table 1.

References

[1] M. Kauers and C. Koutschan, *Some D-finite and some possibly D-finite sequences in the OEIS*, Journal of Integer Sequences **26** (2023), Article 23.4.5, Section 6.5, Conjecture 20. Published article; arXiv:2303.02793v2.

[2] R. H. Hardin, OEIS entry A181280, created 10 October 2010; formula contributed by M. Kauers and C. Koutschan, 2 March 2023. https://oeis.org/A181280. Retrieved 25 October 2026.